

Visualizing Spacecraft Magnetic Fields on the Web and in VR

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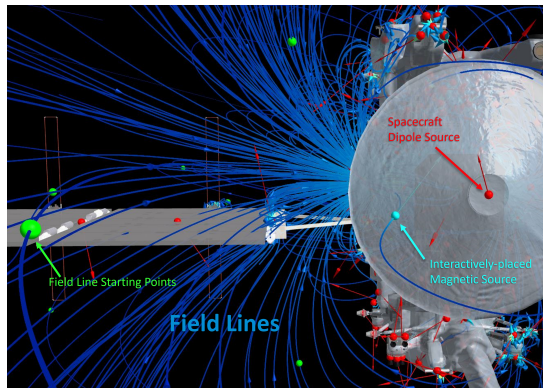
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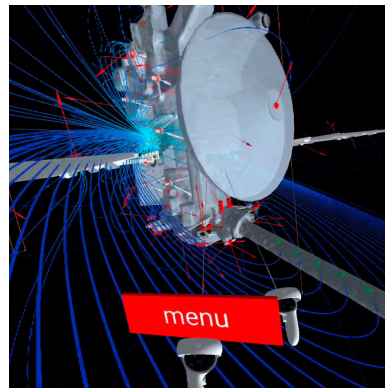
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(a) Description of visualization; see Section 2 for details.



(b) View from the VR experience.



(c) User testing the VR UI.

Figure 1: Views of the prototype tool showing the Europa Clipper spacecraft magnetic field visualized with WebXR.

ABSTRACT

Spacecraft magnetic fields are very complex in nature, and they must be understood and minimized, as they can mask or even mimic signals of interest. Current approaches simulate the 3D nature of the magnetic field and then visualize it in 2D images. However, limited 2D views hide much of the information in the complex spacecraft magnetic field. We describe a prototype system that allows for both an interactive 3D web experience as well as an immersive virtual reality (VR) experience to view and manipulate the 3D spacecraft magnetic field, allowing engineers and scientists to trace field lines, real-time in 3D. A preliminary user study validates the usefulness of the tool and guides further development.

CCS CONCEPTS

• **Applied computing** → **Physical sciences and engineering**; • **Human-centered computing** → **Interaction paradigms**.

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1 INTRODUCTION

Modeling and understanding spacecraft magnetic fields is extremely important, especially when planetary science instruments, such as magnetometers and plasma sensors, must not be contaminated by spacecraft magnetic fields in order to receive accurate measurements and achieve science goals. To achieve magnetic cleanliness, electromagnetic compatibility (EMC) engineers and scientists characterize the individual magnetic sources of the spacecraft, model and simulate the resulting overall magnetic field of the spacecraft [3], and potentially add compensation magnets or provide magnetic shielding appropriately [4]. Visualization of the simulated magnetic field is an important step in this process; however, currently visualizations are generated in a slow, non-real-time way through Matlab scripts. Having the ability to both run the field line

tracing algorithm in real-time and visualize and interact with it has the potential to help engineers achieve magnetic cleanliness goals faster and with more confidence. Furthermore, being able to adjust various parameters of the field line algorithm and to interactively add magnetic sources to the simulation can help EMC engineers understand if compensation magnets can help reduce any spacecraft magnetic field contamination of vital sensors.

In the past, augmented reality (AR) and VR have been used to visualize magnetic fields [11, 16]. Recent use cases include education [7, 9], safety [13], and intuitive understanding of the field [5]. However, we have not seen any attempts yet to visualize a *spacecraft* magnetic field in an immersive way.

The contributions of this work are: (1) to our awareness, the first-ever VR spacecraft magnetic field visualization tool; and (2) the results of a preliminary user study that validates the usefulness of the tool and provides direction for future work.

2 PROTOTYPE TOOL

To understand what features are most important, we conducted a series of informal interviews with EMC engineers and electromagnetic subject matter experts (SMEs). We then iteratively built an initial prototype, gaining feedback along the way with the SMEs. The prototype was built using WebXR with three.js [15] and React [8]. We particularly chose WebXR so that the tool can easily be used either on a standard computer or with an AR/VR headset.

As shown in Figure 1a, the tool visualizes various things. First, the green spheres are the starting points for the magnetic field generation algorithm; we use a numerical method for tracing out lines based on the Euler Method [3]. Initial experiments indicate our Javascript conversion of the Matlab code has around a 10x speedup (e.g., 100 seconds down to 10 seconds). The bluish lines are the magnetic field lines, where the field lines' color indicates the strength (magnitude) of the magnetic field, using a variation of a "jet" color map [14]. Note that unlike traditional field line visualizations, the density of the field lines does not necessarily indicate strength of the magnetic field (as already mentioned, strength is indicated by color). The field lines have arrows (3D cones) to indicate directionality. The red spheres and arrows are the spacecraft's dipole sources' locations and directions, where the length of the arrow is logarithmically proportional to the strength of the dipole source moment. The teal spheres and arrows are any interactively-placed magnetic sources the user placed using our tool. Finally, the tool currently supports two spacecraft, Europa Clipper [1, 12] and Psyche [6, 10].

For scene navigation, the web UI uses traditional orbit controls, while the VR UI uses a bimanual approach to rotate and move the scene. When both controller triggers are pressed, an initial vector offset between the controllers' positions is calculated; then the scene is rotated by calculating the rotation difference between that initial offset and the instantaneous controllers' positions offset. VR movement is towards or away from the midpoint of the intersections of the controllers' rays with the spacecraft when the controller triggers are initially pressed; as the controllers move towards each other, the user moves towards that midpoint, and vice versa.

A large variety of controls are available, including toggling visibility of spacecraft, field lines, and magnetic sources; adjusting field line sizes; randomization of moment orientations; starting

point selection; and ambient field contribution (added after the user study). Users can place new magnetic sources and run the simulation again, with starting points being added next to each newly added source. A VR menu (shown collapsed in Figure 1b) has a subset of visibility toggles, ability to run the simulation, and the ability to add magnetic sources. Adding sources is achieved by raycasting the mouse pointer or controller rays with the spacecraft CAD model; pressing the mouse or controller trigger buttons add sources to the scene; users can also move or delete the sources.

Preliminary testing was done primarily on Windows and Mac laptops and Meta Quest 2 headsets. Quest 2 frame rate typically achieves around 45 FPS, which varies mostly depending on how many field lines (and points on each line) are currently visualized as well as the complexity of the spacecraft CAD model (we reduce CAD complexity using quadric edge collapse in Meshlab [2]).

3 PRELIMINARY USER STUDY

A preliminary user study was conducted with 8 participants (7 male; avg. 33 years old) to understand if and when the tool brings value to end users. Four participants stated having over 10 years of experience working with magnetic fields, three with 2-5 years, and one 0-2 years. All stated 0-2 years of VR experience.

Participants first filled out a pre-study questionnaire and then experienced the web UI on a Windows laptop and the VR UI in a Quest 2, in sequence, with the order being counterbalanced among all eight participants. Finally, a post-study questionnaire was filled out. The entire process took around 45 to 60 minutes.

We asked users about existing and potential new features for the tool, including "Web Browser Viewing," "VR Viewing," "Scene navigation," "Field generation controls (e.g., starting points, dipole randomization, etc.)," "Interactive magnetic source placement," "Ability to save/load/compare simulations," "Viewing simulations with others at the same time," "Exporting videos/screenshots/data from simulations," and "Augmented Reality viewing (e.g., in a cleanroom)." The top features reported as "Extremely valuable" were "Web Browser viewing" (7 stated this); "Exporting videos/screenshots/data from simulations" (7 stated this); "Scene navigation" (6 stated this); and "Ability to save/load/compare simulations" (6 stated this).

When asked about "real-world electromagnetic tasks," five participants "strongly agree[d]" that the web UI has strong potential to be useful, versus only three for the VR UI. When asked about "public engagement," five "strongly agree[d]" that the web UI has strong potential to be useful, versus seven for the VR UI. One participant stated, "This will be very useful to visualize the magnetic field interactions from [an] engineering perspective. We can then identify real world locations on how to mount or implement magnets. For public outreach, this gives people a chance to see magnetic fields and what it is doing, that can't be seen by the naked eye."

We asked participants to state their overall preference of UI, with three choices being "only web", "only VR", or "both web & VR." Five preferred "both web & VR," followed by two preferring "only web," and one preferring "only VR." Comments indicated users felt that the web UI has better precision and accessibility, while the VR UI gives a better "real-life view of the model"; another stated, "VR helps dramatically for seeing exactly where to implement things."

4 CONCLUSION

We presented a 3D spacecraft magnetic field visualization tool that allows EMC engineers and scientists to understand and interact with the magnetic field. User experiment results indicate the usefulness of having both a web and VR UI, and the strong potential usefulness of the tool. Future work will focus on fine-tuning the tool for EMC teams usage and integrating with their workflow.

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